BEATRISE BĒRTULE

Master's Cybersecurity Student

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SUMMARY

Cybersecurity MSc student at Radboud University with research experience in deep-learning side-channel attacks on cryptographic hardware and hands-on offensive skills from CTF competitions. Software development background in Java, Python and Docker. Open to roles across offensive and defensive security, with a particular interest in cryptographic engineering, hardware security and privacy-enhancing technologies.

SKILLS

Languages: Python, C, C++, Java, Haskell

Technologies: PyTorch, TensorFlow, Kali Linux, Git, Linux/Unix, Docker

PROJECTS

Python, PyTorch — **Deep-Learning Side-Channel Attacks (BSc Thesis)** — **github.com/beatrisebertule**
Recovered secret keys from cryptographic devices with Mamba neural network adapted for side-channel attack context and trained on protected AES128 power-consumption traces. Thesis title: Mamba for Side-Channel Attacks:Bidirectional State-Space Encoder Against Masked AES.

Java, Java Card — **Smart-Card Security Protocol** — **github.com/beatrisebertule**
Designed and implemented a smart-card vehicle rental system with a Java Card applet and terminals. Built a custom security protocol with ECDSA P-256 mutual authentication, certificates, nonce-bound PIN verification and replay protection, validated by 245 tests including malicious-terminal and malicious-card attacks.

Python, Docker, LLMs — **Indirect Prompt Injection Defenses** — **github.com/beatrisebertule**
Implemented spotlighting defenses against indirect prompt injection in LLM agents (delimiting, datamarking and encoding, with randomized variants). Evaluated them on the BIPIA, InjecAgent and AgentDojo benchmarks, adapted to run on locally hosted models and DeepSeek.

Python, Streamlit — **Organic Chemistry Learning Platform** — **github.com/beatrisebertule**
Web-based platform where undergraduate chemistry students practise organic molecule structures, containerised with Docker.

Java, Timefold, React — **Warehouse Space Optimiser**
Solved a warehouse bin-packing optimisation problem with the Timefold solver in Spring Boot, with an interactive React frontend to visualise solutions.

CTF — **CTF Write-ups**
Detailed write-ups of challenges solved with the Radboud University CTF group, documenting solutions and methodology across multiple categories.

EDUCATION

02/2026 - present — **MSc Cybersecurity** — **Radboud University, Nijmegen**
Radboud CTF Group.

09/2022 - 01/2026 — **BSc Computer Science** — **Radboud University, Nijmegen**
Cybersecurity track, AI minor.
Thesis on deep-learning side-channel attacks.

EXPERIENCE

09/2025 – 01/2026 — **Research Intern (BSc Thesis)** — **Digital Security Lab, Radboud University**

- Adapted Mamba (state-space) neural network for side-channel attacks on protected AES128.
- Preprocessed power consumption traces, analyzed model performance and optimized attack strategies; documented the work for reproducibility.

`Python / PyTorch / TensorFlow`

07/2025 – 02/2026 — **Freelance Software Developer** — **Faculty of Science, Radboud University**

- Developed a web-base platform to enable undergraduate chemistry students to practice and explore organic molecule structures.
- Containerized the application and shipped features as part of a development team.

`Python / Streamlit / Docker`

02/2025 – 07/2025 **Software Developer Intern** **XLRIT Solutions**

- Solved a warehouse bin-packing optimisation problem with the Timefold solver.
- Built an interactive frontend to visualise solutions and deployed it with Docker.

`Java / Spring Boot / React / Docker`

09/2023 – present **Student Assistant** **Faculty of Science, Radboud University**

- Support students one-on-one in practical sessions.
- Grade assignments against rubrics and give lecturers feedback on lesson flow and engagement.

`Teaching / Mentoring`

09/2022 – present **Student Ambassador & Administrative Assistant** **International Student Office, Radboud University**

- Guide prospective international students through admissions and explain policies clearly to people from diverse backgrounds.

`Communication`

LANGUAGES

Latvian - native, **English** - C1, **Dutch** - A1